



Biology Chapter 8: Part 4

Gas Exchange aur Homeostasis (8.3, 8.4)

Aasan Notes, tumhari kitab ke saath saath (book pages 121 se 125)

Assalam o alaikum Seemab! Ye chapter 8 ka **aakhri hissa** hai, Part 4 of 4. Is mein do badi cheezein hain: **8.3 Gas Exchange in Plants** aur **8.4 Homeostasis in Plants**. Ye notes kitab ki **har line** ke saath chalte hain, is liye kitab kholne ki zaroorat nahin. Headings aur page numbers tumhari apni kitab wale hi hain.

Kaise parhna hai (taqreeban 30 minute):

1. Ek dafa aaram se parho. Ratna nahin, samajhna hai.
2. Phir sirf **neelay**  **boxes** dobara parho. Yehi wali English lines exam mein likhni hain.
3. Aakhir mein **Exercise ke hal shuda sawal** dekho, aur phir **"Khud ko check karo"** kar lo.

Part 4 ka pehla lafz kitab ke page 121 ke neeche "8.3 GAS EXCHANGE IN PLANTS" hai, aur aakhri lafz page 125 par osmotic adjustment ke point 5 (succulents) par khatam hota hai. Us ke baad page 126 par SUMMARY shuru ho jati hai.

 **Student Learning Outcomes (kitab ke chapter opener par, SLOs 11 se 14): Part 4 ke baad tumhein ye aana chahiye**

- 11. Describe the process of gaseous exchange in plants.
- 12. Define homeostasis and describe its importance.
- 13. Describe the mechanism of adaptations of plants for the excretion.
- 14. Explain osmotic adjustment in plants.

Ye chaaron neeche notes mein poore aa rahe hain, aur teenon Section III ke lambe sawal (Q7, Q8, Q9) inhi chaar SLOs se bante hain.

Part 4 ka naqsha (4 hisse)

 Hissa 1	Gas exchange: patton mein aur jaron mein (8.3, 8.3.1, 8.3.2)
 Hissa 2	STEAM Activity: Green Breathing (p123, p124)
 Hissa 3	Homeostasis aur excretion (8.4, 8.4.1, 8.4.2)
 Hissa 4	Osmotic adjustment aur teen plant groups (8.4.3)



Hissa 1: Gas Exchange in Plants

8.3 Gas Exchange in Plants (kitab p121)

Pehle ek asaan baat. Hum saans lete hain to **ek hi jagah** se, naak se. Paudhe ke paas naak nahin hoti. Paudhe ka **har cell apni gas exchange khud** karta hai, apni zaroorat ke mutabiq. Koi pump nahin, koi lungs nahin, bas gas khud ba khud zyada wali jagah se kam wali jagah ki taraf chali jati hai. Isi ko **diffusion** kehte hain.



Kitab ke alfaaz: exam mein yehi likho

"Gas exchange is the type of respiration that occurs at organismic level. Gas exchange is when the entire organism exchanges gases with the environment."

Kitab is ka **maqsad** bhi batati hai: "The goal is to make sure that the organism must be provided with **oxygen** which is needed for **cellular respiration**, where complex organic compounds are broken down, and **carbon dioxide** is produced which is to be removed in outer environment."

- **"Every cell in a plant does its own gas exchange**, depending on what it needs."
- Ye exchange khaas tor par **do openings** se hota hai: **stomata** (tiny openings in leaves, patton ke chhote chhote sooraakh) aur **lenticels** (small openings in stems, tanon ke chhote sooraakh).
- "This exchange happens through a process called **diffusion**."

Do lafz yaad rakhne ka tareeqa

Stomata mein "S" hai: **Subz patta** (leaf). **Lenticels** mein "L" hai: **Lakri wala tana** (stem). Bas.

Ek line mein

Gas exchange = poore organism ka mahol ke saath gas ka len den, organismic level ki respiration. Har cell apna kaam khud karta hai, stomata (leaves) aur lenticels (stems) ke zariye, diffusion se.

8.3.1 Pattern of gas exchange in leaves (Photosynthetic parts)

(kitab p122)

Kitab: "The plants being **photosynthetic autotrophs** show **different pattern** of gas exchange in day and night through the leaves."

Matlab: patte din mein ek tarah se gas lete dete hain, aur raat mein bilkul ulta.

Table 8.1: Pattern of gas exchange between plant and environment
(poori table, sirf 2 lines hain, zaroor yaad karo):

	In daytime	In nighttime
Inhale (andar lena)	Carbon dioxide (CO ₂)	Oxygen (O ₂)
Exhale (bahar chhorna)	Oxygen (O ₂)	Carbon dioxide (CO ₂)

Din mein kya hota hai. Kitab: "During the day, plants are busy with **both photosynthesis (making food) and respiration**. The rate of photosynthesis varies throughout the day as it mainly depends upon **light intensity**. Generally, the rate of photosynthesis is **greater than** rate of respiration, therefore, the photosynthesis needs **more carbon dioxide** than what respiration produces, so plants bring extra carbon dioxide from the environment. On the other hand, photosynthesis produces **more oxygen** than respiration needs, so plants release excess oxygen."

Ghar ki misal

Socho ghar mein **roti ban rahi hai** aur saath saath **khai bhi ja rahi hai**. Din mein banane ki raftaar khane se zyada hai, is liye aata (CO_2) bazaar se mangwana parta hai, aur bachi hui roti (O_2) bahar bhej di jati hai. Raat mein banana band, sirf khana chalta hai.

Sab se bara ghalat fehmi

Bohat log kehte hain "din mein paudha sirf photosynthesis karta hai aur raat mein sirf respiration". **Ghalat**. Respiration **chobees ghante** chalti rehti hai. Din mein sirf photosynthesis ki **raftaar zyada** hoti hai, is liye net mein CO_2 andar aur O_2 bahar nazar aata hai.

Figure 8.10: Pattern of gas exchange in leaves and roots in day and night. Tasveer mein ek darakht hai. Upar ki taraf "In day" par CO_2 andar aur O_2 bahar ke teer hain, aur "At night" par O_2 andar aur CO_2 bahar. Zameen ke neeche jaron par **dono waqt** ek jaisa hai: O_2 andar, CO_2 bahar.

Raat mein kya hota hai. Kitab: "At night, plants act more like **animals**. They **absorb oxygen and release carbon dioxide** because there's **no photosynthesis** happening in the dark."

Compensation point of photosynthesis (kitab p122)

Ab subah saweray (**dawn**) aur shaam dhalte waqt (**dusk**) ki baat. Us waqt roshni **halki** hoti hai. Photosynthesis ki raftaar gir kar theek respiration ki raftaar ke **barabar** aa jati hai.

Kitab ke alfaaz: exam mein yehi likho

"During dawn and dusk, when light intensity is low, the rates of photosynthesis and respiration become equal. This means the carbon dioxide produced by respiration is enough for photosynthesis, and the oxygen released by photosynthesis is used in respiration. At this point, there's no net exchange of gases with the environment, and we call it the "compensation point of photosynthesis.""



Samajhne ka tareeqa

Jitni CO_2 respiration banati hai, utni hi photosynthesis kha jati hai. Jitni O_2 photosynthesis banati hai, utni hi respiration kha jati hai. **Dono ka hisaab barabar**, is liye mahol ke saath **koi net len den nahin**. Gas **andar hi andar** ghoomti rehti hai.

🎯 Pakki trick: "no net exchange" ka matlab

Compensation point par **na CO_2 absorb hoti hai, na O_2 absorb hoti hai** (mahol se). Ye MCQ 6 ka seedha jawab hai: option **D**. Yaad rakho: "compensation" ka matlab hi hai **barabar kar dena**.

8.3.2 Pattern of gas exchange in roots (non-photosynthetic parts) (kitab p122)



Kitab ke alfaaz

"Roots also exchange gases with soil air through root epidermal cells which involves absorption of oxygen and release carbon dioxide because there's no photosynthesis happening in the roots and it remains same day and night."

- Jarein mitti ki hawa (**soil air**) se gas exchange karti hain.
- Ye kaam **root epidermal cells** karte hain.
- Jarein sirf **respiration** karti hain, photosynthesis bilkul nahin (wahan roshni hi nahin).
- Is liye jarein **hamesha O_2 leti hain aur CO_2 chhorti hain**, din ho ya raat, pattern **badalta nahin**.



Roz ki misal

Gamle mein zaroorat se zyada paani daal do to paudha murjha jata hai. Wajah: mitti ke sooraakh paani se bhar gaye, jaron ko **oxygen** mili hi nahin. Jaron ko bhi saans chahiye.

✨ Ek line mein

Patte: din mein CO_2 in aur O_2 out, raat mein ulta, dawn aur dusk par kuch bhi nahin (compensation point). Jarein: hamesha O_2 in aur CO_2 out, din raat ek jaisa.



Hissa 2: STEAM Activity

Green Breathing: Unveiling Plant Respiratory Patterns in Day and Night (kitab p123, p124)

Objective: "To recognize how does a plant's exposure to light and darkness impact its gas exchange?"



Principle (kitab ke alfaaz)

"The increase or decrease in carbon dioxide concentration will be tested in this inquiry by using hydrogen carbonate indicator. It goes from pink or red to yellow when carbon dioxide builds up. The indicator becomes purple when carbon dioxide concentrations fall."

Indicator ka rang yaad karne ki chhoti table:

CO ₂ ka haal	Indicator ka rang
Shuru mein (normal)	Pink ya red
CO ₂ barh gayi (builds up)	Yellow
CO ₂ kam ho gayi (falls)	Purple

Procedure (kitab ke poore 5 steps):

1. "Take three test tubes of **50ml** volume and wash them properly with tap water then rinse them with distilled water and finally with hydrogen carbonate indicator (the indicator will change colour if the boiling tube is not clean)."
2. "Add **30ml** hydrogen carbonate indicator solution into each of the three boiling tubes."
3. "Place equal-sized pieces of **Canadian pondweed (Elodea)** or **Hydrilla** or any other locally available aquatic plant in **tubes 1 and 2**, then use stoppers to close all of the tubes."
4. "Use a bench lamp to expose **tubes 1 and 3 to light**, then cover **tube 2** with aluminum foil or put it in a dark cabinet or black box (Figure 8.11)."

5. "Note the colour of each tube's hydrogen carbonate indicator after a period of **24 hours**."

Figure 8.11: Experiment to compare gas exchange in plants kept in the dark and in the light. Tasveer mein teen tubes hain. Tube **2** ek kaale dabbe ke andar (andhera) aur us mein pond weed hai. Tube **1** light ke saamne, us mein bhi pond weed hai. Tube **3** light ke saamne magar **khaali** (sirf indicator). Neeche likha hai "Hydrogencarbonate indicator (pink/red at start)".

 Har tube ka kaam

Tube 1 = paudha + roshni (photosynthesis chalegi). **Tube 2** = paudha + andhera (sirf respiration). **Tube 3 = control**, paudha nahin, sirf roshni. Control is liye rakha ke pata chale rang ka farq **paudhe ne** kiya hai, roshni ne nahin.

 Result (kitab ke alfaaz)

"The indicator in tube 2 (plant in the dark) should turn yellow, the indicator in tube 1 (plant in the light) should turn purple, and the indicator in tube 3 (the control), which was initially pink/red, should not change colour."

Tube	Halat	Rang	Kyun
1	Plant + light	Purple	Photosynthesis ne CO ₂ istemal kar li, CO ₂ gir gayi
2	Plant + dark	Yellow	Sirf respiration, CO ₂ barh gayi
3	Control (koi plant nahin)	Koi tabdeeli nahin (pink/red)	Rang badalne wala paudha hi tha

 Interpretation (kitab ke alfaaz, p123 se p124 tak)

"In simple terms, hydrogen carbonate indicator helps us see changes in acidity caused by carbon dioxide. When there's more carbon dioxide, it turns orange/red, and when there's less, it becomes purple. An experiment with this indicator and plants showed that in light, plants use more carbon dioxide in photosynthesis than they produce in respiration. In the dark, plants produce carbon dioxide through respiration. A control tube without a plant showed that the plant caused the color change. However, it's noted that the indicator can react to changes in acidity or alkalinity, not just carbon dioxide. Despite this, understanding how leaves work suggests changes in carbon dioxide are more likely."

Reflection: "Suggest students to emphasize the significance of gas exchange in plants for animal life on Earth. Encourage them to contemplate the potential consequences if plants were to become extinct."

Yani: soch kar dekho, agar paudhe khatam ho jayen to janwaron (aur hum) ko oxygen kahan se milegi?

 Ek line mein

Roshni mein paudha CO₂ khaata hai (indicator purple), andhere mein CO₂ banata hai (indicator yellow), aur khaali control tube ka rang nahin badalta.



Hissa 3: Homeostasis aur Excretion

8.4 Homeostasis in Plants (kitab p124)

Homeostasis ka matlab hai paudhe ka **apne andar ka balance qaim rakhna**, chahe bahar kuch bhi ho raha ho. Garmi bohat ho, paani kam ho, mitti mein namak zyada ho, paudha apne aap ko us ke mutabiq adjust kar leta hai.



Kitab ki definition: exam mein yehi likho

"Homeostasis in plants refers to the adjustment or adaptation of plants to varying degrees of environmental temperatures and availability of water and salts in the soil."

Definition ke teen tukre yaad karo: **(1)** temperature, **(2)** paani ki availability, **(3)** mitti mein salts ki availability.

8.4.1 Importance of Homeostasis (kitab p124)

Kitab ne is ke kayi faide ginwaye hain. Har ek alag point hai, exam mein sab likh sakti ho:

1. "Homeostasis in plants is essential for their **survival, growth, and reproduction.**"
2. "Homeostatic mechanisms help plants **respond to temperature changes**, allowing them to survive in **various climates.**"
3. "Homeostasis ensures that plants maintain **adequate hydration levels.**" (jism mein paani ki sahi miqdar)
4. "Homeostasis allows plants to respond to various **stressors, such as drought, salinity, pathogens, and herbivores.**" (khushk sali, namak, beemari wale jarasim, aur ghaas patta khane wale janwar)
5. "Plants can activate specific **defense mechanisms** to mitigate damage." (nuqsan kam karne ke liye apna difaa chala lete hain)
6. "It enables them to **adapt to changing environmental conditions** and maintain **internal stability**, ultimately contributing to their **ecological**

success."

Chaar stressors yaad karne ka tareeqa

Drought, Salinity, Pathogens, Herbivores. Yaad rakho: "**D S P H**" yani **D**hoop, **S**alt, **P**hupoondi, **H**irni. Chaaron alag alag khatra hain.

8.4.2 Excretion in plants (kitab p124)

Janwaron mein excretion ka matlab hota hai gurdon (kidneys) se fazla nikalna. Paudhon mein baat **bohat wasee** hai: jism se **koi bhi cheez** nikalna excretion hai.

Kitab ke alfaaz

"Unlike animals, excretion in plants is a very broad term, which encompasses the removal of anything from plant body. Plants have a different way of getting rid of waste compared to animals. Plants make various organic and inorganic compounds that they store for different reasons and remove when needed. Following are different ways of excretion in plants."

Kitab **paanch tareeqe** bayan karti hai. Section III ka Q8 poora yehi hai, is liye paanchon yaad karo:

1. Excretion of carbon dioxide and oxygen (kitab p124)

- "**Carbon dioxide** is the waste of **respiration** which is excreted from plant body **at night** during gas exchange process."
- "Similarly, **oxygen** is the waste of **photosynthesis** which is excreted **in day** during the same gas exchange process."

Dhyan do: paudhe ke liye **oxygen bhi ek waste hai**. Hamare liye zindagi, us ke liye bacha hua maal.

2. Excretion of water (kitab p124)

- "Water is produced as waste of **both photosynthesis and respiration** and it is also **absorbed from soil**."

- "The excretion of water occurs through the plant body during the process of **transpiration**."

3. Excretion through leaves (kitab p124)

Kitab ke alfaaz

"Plant cells have large vacuoles that can store useful stuff or waste. Sometimes, these stored substances can build up and form crystals in the vacuoles. Leaves are key players in this process. When the leaves are loaded with large amount pigmented compounds, they turn yellow. Remember, this yellowing is not due to lack of chlorophyll as happens in chlorosis. Such leaves are generally fallen from plants in autumn season. In this way leaves act as organ of excretion, therefore, such leaves are also called excretophores. This is why gardeners like using decomposed autumn leaves as a mineral-rich source for plants."

Asaan alfaaz mein:

- Plant cells ke **bare vacuoles** hote hain. In mein kaam ki cheez bhi rakhi ja sakti hai aur fazla bhi.
- Kabhi kabhi ye jama shuda cheezein **crystals** bana leti hain.
- Jab patte **pigmented compounds** se bhar jate hain to **peelay** ho jate hain.
- **Ye peelapan chlorosis nahin hai.** Chlorosis mein chlorophyll ki kami hoti hai; yahan chlorophyll ki kami nahin, balke fazla jama ho gaya hai.
- Aise patte **autumn (khizan)** mein gir jate hain. Patta gira, fazla bhi saath gaya. Is liye patte ko **organ of excretion** kehte hain aur aise patton ko **excretophores**.
- Isi wajah se maali (**gardeners**) **decomposed autumn leaves** ko mitti mein daalte hain: ye **mineral-rich source** hain.

? **Critical Thinking box (kitab p124): "Can you differentiate between chlorotated leaves and excretophores?"**

Chlorotic leaves: peelay is liye hain ke **chlorophyll ban hi nahin paya** (mineral ki kami, jaise magnesium ya iron). Ye **beemari** ki nishani hai.

Excretophores: peelay is liye hain ke un mein **pigmented waste compounds bhar gaye**. Chlorophyll ki koi kami nahin. Ye **normal** hai aur ye patte autumn mein gir kar paudhe ka fazla saath le jate hain.

Kitab mein lafz "chlorotated" chhapa hai; asal lafz **chlorotic** hai (chlorosis se). Ye kitab ki printing ki ghalti hai.

4. Excretion through stem and branches (kitab p125)

 **Kitab ke alfaaz**

"Some trees deposit unusual chemicals in their branches and trunks, especially in old xylem that's no longer used for water transport. Ebony trees, for instance, produce very dark wood in their centres. Plant scientists consider these deposits as waste materials. Although, these wastes are not removed but are dumped in one part of the body as we put the trash in waste bin however, it is still present in the home."

- Kuch darakht apne **branches aur trunks** mein ajeeb chemicals jama kar lete hain.
- Khaas tor par us **purane xylem** mein jo ab paani le jane ka kaam nahin karta.
- **Ebony trees** (aabnoos) ki misal: in ke beech ka hissa bohat **gehra kaala (very dark wood)** ho jata hai.
- Plant scientists in deposits ko **waste materials** mante hain.
- Kitab ki misal: ye fazla jism se nikalta nahin, bas ek hissay mein **daal diya jata hai**, jaise ghar ka koora **waste bin** mein daal dete hain magar woh ghar hi mein maujood rehta hai.

5. Excretion through roots (kitab p125)

Kitab ke alfaaz

“Some plants actively release waste compounds into the soil, almost like using them as chemical weapons against other plants competing for resources. Conifer trees are known for this tactic.”

- Kuch paudhe apna fazla **mitti mein chhor dete hain**.
- Ye fazla un ke liye **chemical weapon** ka kaam karta hai un doosre paudhon ke khilaf jo wahan ke resources (paani, minerals) ke liye muqabla kar rahe hain.
- **Conifer trees** is chaal (tactic) ke liye mashhoor hain.

Ek line mein

Excretion ke **paanch** raaste: gases (CO_2 raat ko, O_2 din ko), paani (transpiration se), patte (excretophores), stem aur branches (ebony ki dark wood), aur jarein (conifer ke chemical weapons).



Hissa 4: Osmotic Adjustment

8.4.3 Osmotic adjustment (Osmoregulation) in plants (kitab p125)

 **Kitab ki definition: exam mein yehi likho**

"Osmotic adjustment, also known as osmoregulation, is like a plant's way of maintaining the right balance of water and solutes in its body."

Kitab: "This balance creates **three different situations: hypotonic, hypertonic, and isotonic.**"

Hypotonic Situation (kitab p125)

Kitab: "When a cell is in a watery environment with **more water and less stuff dissolved in it (solute)**, water moves **into the cell**, making it swell up and firm (we call this '**turgid**')."

Bahar paani zyada, solute kam. Paani cell ke **andar** aaya. Cell phool kar **kasa hua (turgid)** ho gaya.

Hypertonic Situation (kitab p125)

Kitab: "If a cell is in an environment with **less water and more dissolved stuff** than it has inside (**cell sap**), water moves **out of the cell**, causing it to shrink and become limp (we call this '**flaccid**')."

Kitab yahan ek zaroori jumla likhti hai: "**Both of these situations, hypotonic and hypertonic, require the plant to make adjustments to keep things in balance.**"

Ideal Situation (Isotonic) (kitab p125)

Kitab: "The best scenario for a cell is when the water and solute balance inside and outside the cell is **just right**. In this '**isotonic**' situation, there's **no**

net movement of water in or out of the cell. **However, this balance is rarely found in a plant's natural environment."**

Situation	Bahar ka mahol	Paani kidhar	Cell ka haal
Hypotonic	Paani zyada, solute kam	Cell ke andar	Turgid (phoola aur kasa hua)
Hypertonic	Paani kam, solute zyada	Cell se bahar	Flaccid (sukra aur dheela)
Isotonic	Dono taraf barabar	Koi net movement nahin	Normal, magar qudrat mein bohat kam milta hai

Hypo aur Hyper ki pakki trick

Hypo mein "o" hai, jaise "**o**ver water". Bahar paani zyada to cell **phoolta** hai (turgid).

Hyper mein "er" hai, jaise "**dry** paani nikal jaye". Bahar solute zyada to cell **sukarta** hai (flaccid).

Aur yaad rakho: **paani hamesha kam solute wali taraf se zyada solute wali taraf jata hai.**

Ghar ki misal

Salad ke patton par **namak** chhirak do: thori der mein patte **dheele aur murjhaye** ho jate hain. Namak ne bahar solute barha diya (hypertonic), paani cells se nikal gaya, cells **flaccid** ho gaye. Ab wohi murjhaya patta **saaf paani** mein daalo: paani wapis andar jayega (hypotonic) aur patta phir **kara (turgid)** ho jayega.

Osmotic adjustment in different plants groups (kitab p125)

📖 Kitab ke alfaaz

“Plants are categorized into three groups based on their water availability: **hydrophytes** (live in water-rich environments), **mesophytes** (thrive where water is moderately available), and **xerophytes** (adapted to extreme dry conditions). These groups have distinct adaptations:”

- **Hydrophytes:** paani se bharay mahol mein rehte hain (jaise lotus, water lily).
- **Mesophytes:** jahan paani **darmiyani** ho (aam paudhe, gamle wale paudhe, fasalein).
- **Xerophytes:** bohat **khushk** jaghon ke liye bane hue (jaise cactus, thor).

Kitab ke poore 5 differences (ye table exam mein bohat aati hai):

#	Farq	Hydrophytes	Mesophytes	Xerophytes
1	Transpiration rate	Highest (sab se zyada)	Darmiyani	Lowest (sab se kam)
2	Stomatal placement	Upper leaf surface par	Lower surface par	Sunken stomata (dhansay hue)
3	Stomatal behaviour	Din raat khulay rehte hain	Sirf din mein khulte hain	Raat mein khulte hain
4	Cuticle thickness	Almost none (taqreeban nahin)	Thinner cuticle (patli)	Thickest (sab se moti)
5	Water storage	Zaroorat hi nahin	Thori si	Ample water , inhein succulents kehte hain

Kitab ke apne alfaaz mein point 1: "Transpiration rates vary, with hydrophytes having the highest and xerophytes the lowest." Point 2: "Stomatal placement differs, hydrophytes have them on the upper leaf surface, mesophytes on the lower surface, and xerophytes have sunken stomata." Point 3: "Stomatal behaviour varies, hydrophytes keep stomata open day and night, mesophytes open them during the day, and xerophytes open them at night." Point 4: "Cuticle thickness on leaves differs, with

xerophytes having the thickest, mesophytes a thinner cuticle, and hydrophytes almost none." Point 5: "Water storage capacity varies, with xerophytes, known as succulents, storing ample water."

📌 **Pakki trick: sab kuch ek soch se**

Sirf ek sawal pucho: "**is paudhe ko paani bachana hai ya nahin?**"

Hydrophyte paani mein khara hai, paani bachane ki zaroorat nahin: stomata upar (taa ke paani ke upar khule rahen), hamesha khulay, cuticle ki zaroorat nahin, transpiration sab se zyada.

Xerophyte ko har boond bachani hai: stomata andar dhansay hue, sirf **thandi raat** mein khulte hain, cuticle sab se moti, transpiration sab se kam, aur paani jama kar leta hai (succulent).

Mesophyte beech mein hai: har cheez darmiyani.

✨ **Ek line mein**

Hypotonic = turgid, hypertonic = flaccid, isotonic = koi net movement nahin (qudrat mein bohat kam). Hydrophytes paani mein, mesophytes darmiyani, xerophytes khushki mein aur ye succulents kehlate hain.



Exercise ke hal shuda sawal (Part 4 wale)

Ye woh sawal hain jo kitab ke Exercise (p126 se p128) mein Part 4 ke hisse se bante hain. Har jawab woh hai jo tum **exam mein poora likh sakti ho**.

Section I: Multiple Choice Questions

MCQ 4. Which category of plants stores a small amount of water and has a thin cuticle?

A) Hydrophytes B) Xerophytes C) Mesophytes D) Succulents

Jawab: C) Mesophytes

Wajah (kitab ke points 4 aur 5 se): Cuticle ke bare mein kitab kehti hai xerophytes ki sab se **moti**, mesophytes ki **thinner (patli)**, aur hydrophytes ki **almost none**. Is liye "thin cuticle" sirf mesophytes par fit baithta hai. Water storage mein kitab kehti hai **xerophytes** (jo succulents kehlate hain) **ample water** jama karte hain, is liye "small amount of water" bhi mesophytes hi hai.

Baqi options kyun ghalat: A) Hydrophytes: cuticle **almost none** hai, "thin" nahin, aur woh paani hi mein khare hain. B) Xerophytes: cuticle sab se **moti** aur paani **zyada** jama karte hain. D) Succulents: kitab ke mutabiq succulents **xerophytes ka hi doosra naam** hai, is liye B aur D ek hi cheez hain aur dono ghalat hain.

MCQ 5. What is a key role of leaves in managing waste in plants?

A) Storing waste materials B) Producing chlorophyll C) Absorbing water from the soil D) Converting waste into energy

Jawab: A) Storing waste materials

Wajah: Kitab (Excretion through leaves): "Plant cells have **large vacuoles that can store useful stuff or waste... Leaves are key players in this process.**" Patte fazla apne vacuoles mein jama karte hain, phir autumn mein gir kar us fazle ko saath le jate hain, is liye inhein **excretophores** kehte hain.

Baqi options kyun ghalat: B) Chlorophyll banana photosynthesis ka kaam hai, waste management ka nahin. C) Mitti se paani **jarein** (root hairs) absorb karti hain, patte nahin. D) Waste ko energy mein badalne ka zikr kitab mein kahin nahin.

MCQ 6. When the rate of photosynthesis become equal to that rate of respiration in the plant body, which of the following pattern of gaseous exchange occurs between plant and its environment:

A) Carbon dioxide is absorbed, and oxygen is released

B) Oxygen is absorbed, and carbon dioxide is released

C) Both carbon dioxide and oxygen are absorbed

D) Neither carbon dioxide nor oxygen are absorbed

Jawab: D) Neither carbon dioxide nor oxygen are absorbed

Wajah: Ye **compensation point of photosynthesis** hai. Kitab: "the rates of photosynthesis and respiration become equal... At this point, there's **no net exchange of gases** with the environment." Respiration ki bani hui CO_2 photosynthesis kha leti hai, aur photosynthesis ki bani hui O_2 respiration kha leti hai. Mahol se kuch bhi lena nahin parta.

Baqi options kyun ghalat: A) ye **din** ka pattern hai. B) ye **raat** ka pattern hai (aur jaron ka har waqt). C) aisa kabhi nahin hota.

Section II: Short Answer Questions

Q3. Why gardeners like to use autumn leaves in their soil?

Paudhon ke cells ke **bare vacuoles** hote hain jin mein fazla jama hota hai aur kabhi kabhi **crystals** ban jate hain. Jab patte **pigmented compounds** se bhar jate hain to peelay ho jate hain aur **autumn (khizan)** ke mausam mein gir jate hain. Aise patte excretion ka organ hote hain, inhein **excretophores** kehte hain.

In gire hue patton mein paudhe ke jama kiye hue **organic aur inorganic compounds** maujood hote hain. Jab ye patte mitti mein **decompose (gal sarr)** jate hain to un ke minerals mitti mein shamil ho jate hain.

Is liye gardeners decomposed autumn leaves ko mitti mein daalte hain, kyunke ye paudhon ke liye ek "mineral-rich source" hain.

Q5. Write any three osmotic adjustments in hydrophytes.

Hydrophytes woh paudhe hain jo **water-rich environments** mein rehte hain. Un ke osmotic adjustments (koi bhi teen likh do):

1. **Transpiration rate sab se zyada (highest)** hoti hai, kyunke paani ki kami ka koi khatra nahin.
2. **Stomata upper leaf surface** par hote hain (patte ki upar wali satah par), taa ke wo hawa ke saamne rahen.
3. **Stomata din aur raat dono waqt khulay** rehte hain.
4. **Cuticle taqreeban hoti hi nahin (almost none)**, kyunke paani rokne ki zaroorat nahin.

Paanchwan point (water storage) kitab mein khaas tor par xerophytes ke liye likha hai, hydrophytes ke liye nahin. Is liye upar wale chaar mein se koi teen chuno.

Q6. What is the pattern of gas exchange between plant and environment at the time of dawn and dusk?

Dawn (subah saweray) aur dusk (shaam dhalte) ke waqt **light intensity kam** hoti hai. Is wajah se **photosynthesis aur respiration ki rates barabar** ho jati hain.

- Respiration se bani hui **carbon dioxide** theek utni hi hoti hai jitni photosynthesis ko chahiye.
- Photosynthesis se nikli hui **oxygen** theek utni hi hoti hai jitni respiration istemal kar leti hai.

Nateeja: mahol ke saath gas ka koi net exchange nahin hota. Na CO₂ andar aati hai, na O₂ bahar jati hai. Is halat ko "compensation point of photosynthesis" kehte hain.

Q7. (Kitab mein chhapa hai: "White plants absorb carbon dioxide and release oxygen during daytime.")

Kitab ki printing ki ghalti

Yahan "White" nahin, "**Why**" hona chahiye. Sahi sawal: "**Why plants absorb carbon dioxide and release oxygen during daytime?**"

Exam mein bhi isi ka jawab likhna hai.

Din ke waqt **roshni maujood** hoti hai, is liye paudha **photosynthesis aur respiration dono** kar raha hota hai. Photosynthesis ki rate mainly **light intensity** par depend karti hai.

- Aam tor par din mein **photosynthesis ki rate, respiration ki rate se zyada** hoti hai.
- Photosynthesis ko jitni **carbon dioxide** chahiye hoti hai, respiration us se **kam** banati hai. Kami poori karne ke liye paudha **mahol se extra CO₂ andar leta hai**.
- Photosynthesis jitni **oxygen** banati hai, respiration ko us se **kam** chahiye hoti hai. Bachi hui O₂ paudha **bahar chhor deta hai**.

Isi liye din mein paudha carbon dioxide absorb karta hai aur oxygen release karta hai.

Q8 (i). Define: Osmotic adjustment

Osmotic adjustment, jise **osmoregulation** bhi kehte hain, paudhe ka woh tareeqa hai jis se woh apne jism mein **paani aur solutes ka sahi balance** qaim rakhta hai.

English mein: "Osmotic adjustment, also known as osmoregulation, is like a plant's way of maintaining the right balance of water and solutes in its body." Is balance se teen halatein banti hain: **hypotonic, hypertonic aur isotonic**.

Q8 (v). Define: Excretophores

Jab patte **pigmented compounds** (fazla) ki bari miqdar se bhar jate hain to woh **peelay** ho jate hain aur **autumn** ke mausam mein paudhe se gir jate hain. Is tarah patte **organ of excretion** ka kaam karte hain.

Aise patton ko excretophores kehte hain.

Yaad rakho: ye peelapan **chlorophyll ki kami** (chlorosis) ki wajah se nahin hota.

Q9 (b). Write the difference between: Hydrophytes and xerophytes

#	Hydrophytes	Xerophytes
1	Water-rich environment mein rehte hain	Extreme dry conditions ke liye adapted hain
2	Transpiration rate sab se zyada (highest)	Transpiration rate sab se kam (lowest)
3	Stomata upper leaf surface par hote hain	Stomata sunken (andar dhansay hue) hote hain
4	Stomata din aur raat dono khulay rehte hain	Stomata raat mein khulte hain
5	Cuticle taqreeban hoti hi nahin	Cuticle sab se moti (thickest) hoti hai
6	Paani jama karne ki zaroorat nahin	Ample water jama karte hain, inhein succulents kehte hain

Q9 (g). Write the difference between: Pattern of gas exchange in leaves and roots

#	Leaves (photosynthetic parts)	Roots (non-photosynthetic parts)
1	Gas exchange stomata ke zariye hota hai	Gas exchange root epidermal cells ke zariye hota hai
2	Exchange mahol ki hawa ke saath hota hai	Exchange soil air (mitti ki hawa) ke saath hota hai
3	Yahan photosynthesis aur respiration dono hoti hain	Yahan sirf respiration hoti hai, photosynthesis bilkul nahin
4	Pattern din aur raat mein badalta hai	Pattern din raat ek jaisa rehta hai
5	Din: CO ₂ inhale, O ₂ exhale. Raat: O ₂ inhale, CO ₂ exhale	Hamesha: O ₂ absorb, CO ₂ release
6	Dawn aur dusk par compensation point aata hai (koi net exchange nahin)	Compensation point kabhi nahin aata

Section III: Extensive Answer Questions

Q7. Describe the process of gaseous exchange in plants.

Gas exchange kya hai

Gas exchange woh respiration hai jo **organismic level** par hoti hai, yani poora organism mahol ke saath gases ka len den karta hai. Maqsad ye hai ke organism ko **oxygen** mile jo **cellular respiration** ke liye chahiye (jahan complex organic compounds toote hain), aur jo **carbon dioxide** banti hai woh bahar nikal jaye.

Kahan aur kaise

Paudhe ka **har cell apni gas exchange khud** karta hai, apni zaroorat ke mutabiq. Ye exchange khaas tor par do openings se hota hai: **stomata** (patton ke chhote sooraakh) aur **lenticels** (tanon ke chhote sooraakh). Ye sab **diffusion** ke amal se hota hai.

1. Patton mein pattern (photosynthetic parts)

Paudhe photosynthetic autotrophs hain, is liye patton mein din aur raat ka pattern alag hota hai.

	In daytime	In nighttime
Inhale	Carbon dioxide (CO ₂)	Oxygen (O ₂)
Exhale	Oxygen (O ₂)	Carbon dioxide (CO ₂)

Din mein: paudha photosynthesis aur respiration dono kar raha hota hai. Photosynthesis ki rate light intensity par depend karti hai aur aam tor par respiration se **zyada** hoti hai. Is liye photosynthesis ko respiration se milne wali CO₂ kaafi nahin hoti aur paudha mahol se **extra CO₂** leta hai. Doosri taraf photosynthesis respiration ki zaroorat se **zyada O₂** banati hai, is liye paudha fazil O₂ **release** kar deta hai.

Raat mein: andhere mein photosynthesis hoti hi nahin, is liye paudhe **janwaron jaise** behave karte hain: O₂ absorb aur CO₂ release.

Dawn aur dusk par: light intensity kam hone se photosynthesis aur respiration ki rates **barabar** ho jati hain. Respiration ki CO₂

photosynthesis ke liye kaafi hoti hai aur photosynthesis ki O_2 respiration istemal kar leti hai. Mahol ke saath **koi net exchange nahin** hota, aur isay **compensation point of photosynthesis** kehte hain.

2. Jaron mein pattern (non-photosynthetic parts)

Jarein **soil air** ke saath gas exchange karti hain, **root epidermal cells** ke zariye. Wo **oxygen absorb** karti hain aur **carbon dioxide release** karti hain, kyunke jaron mein photosynthesis bilkul nahin hoti. Ye pattern **din aur raat, dono waqt ek jaisa** rehta hai.

Q8. Describe the mechanism of adaptations of plants for the excretion.

Paudhon mein excretion kya hai

Janwaron ke bar aks, paudhon mein excretion ek **bohat wasee (very broad)** lafz hai jis ka matlab hai **paudhe ke jism se kisi bhi cheez ka nikalna**. Paudhon ka fazla nikalne ka tareeqa janwaron se alag hai.

Paudhe tarah tarah ke **organic aur inorganic compounds** banate hain, unhein mukhtalif wajuhat se **jama** karte hain, aur zaroorat par nikal dete hain. Kitab ke mutabiq excretion ke **paanch** raaste hain:

1. Excretion of carbon dioxide and oxygen

Carbon dioxide respiration ka waste hai, jo **raat** ke waqt gas exchange ke amal mein jism se bahar nikaali jati hai. Isi tarah **oxygen** photosynthesis ka waste hai, jo **din** mein usi gas exchange ke amal se bahar nikaali jati hai.

2. Excretion of water

Paani photosynthesis aur respiration **dono** ka waste hai, aur ye **mitti se absorb** bhi hota hai. Paani ka excretion paudhe ke jism se **transpiration** ke amal ke doran hota hai.

3. Excretion through leaves

Plant cells ke **bare vacuoles** hote hain jo kaam ki cheezein ya waste jama kar sakte hain. Kabhi kabhi ye jama shuda cheezein barh kar vacuoles mein **crystals** bana leti hain. Is amal mein **patte sab se aham** hain. Jab patte **pigmented compounds** ki bari miqdar se bhar jate hain to woh **peelay** ho jate hain. Ye peelapan **chlorosis** ki tarah chlorophyll ki kami se nahin hota. Aise patte aam tor par **autumn** ke mausam mein gir jate hain. Is tarah patte **organ of excretion** ban jate hain, is liye aise patton ko **excretophores** kehte hain. Yehi wajah hai ke **gardeners decomposed autumn leaves** ko paudhon ke liye **mineral-rich source** ke tor par istemal karna pasand karte hain.

4. Excretion through stem and branches

Kuch darakht apni **branches aur trunks** mein ghair maamooli chemicals jama kar lete hain, khaas tor par us **purane xylem** mein jo ab paani le jane ke kaam nahin aata. Misal ke tor par **ebony trees** apne beech wale

hisse mein bohat **gehri kaali lakri (very dark wood)** banate hain. Plant scientists in deposits ko **waste materials** samajhte hain. Ye waste jism se nikalta nahin, bas jism ke **ek hisse mein daal diya** jata hai, bilkul jaise hum koora **waste bin** mein daal dete hain magar woh ghar hi mein maujood rehta hai.

5. Excretion through roots

Kuch paudhe apne waste compounds ko **mitti mein chhor dete hain**, taqreeban is tarah jaise unhein **chemical weapons** ke tor par un doosre paudhon ke khilaf istemal kar rahe hon jo resources ke liye muqabla kar rahe hain. **Conifer trees** is chaal ke liye mashhoor hain.

Q9. Explain osmotic adjustment in plants.

Tareef

Osmotic adjustment, jise **osmoregulation** bhi kehte hain, paudhe ka apne jism mein **paani aur solutes ka sahi balance** qaim rakhne ka tareeqa hai. Is balance se **teen halatein** banti hain: **hypotonic, hypertonic aur isotonic**.

1. Hypotonic situation

Jab cell aise paani wale mahol mein ho jahan **paani zyada aur solute kam** ho, to paani **cell ke andar** chala jata hai. Cell phool kar **sakht aur kasa hua** ho jata hai, is halat ko **turgid** kehte hain.

2. Hypertonic situation

Agar cell aise mahol mein ho jahan cell sap ke muqable mein **paani kam aur dissolved stuff zyada** ho, to paani **cell se bahar** nikal jata hai. Cell sukar kar **dheela** par jata hai, is halat ko **flaccid** kehte hain. Ye **dono halatein (hypotonic aur hypertonic)** paudhe se taqaza karti hain ke woh **adjustments** kare taa ke cheezein balance mein rahen.

3. Ideal situation (Isotonic)

Cell ke liye behtareen soorat ye hai ke cell ke **andar aur bahar paani aur solute ka balance bilkul theek** ho. Is **isotonic** halat mein cell ke andar ya bahar paani ki **koi net movement nahin** hoti. Lekin ye balance paudhe ke **qudrati mahol mein bohat kam** milta hai.

Osmotic adjustment in different plants groups

Paani ki availability ke lihaz se paudhon ke **teen groups** hain:

hydrophytes (paani se bharay mahol mein rehne wale), **mesophytes** (jahan paani darmiyani miqdar mein ho) aur **xerophytes** (bohat khushk halat ke liye adapted). In ki adaptations alag alag hain:

Farq	Hydrophytes	Mesophytes	Xerophytes
Transpiration rate	Highest	Darmiyani	Lowest
Stomatal placement	Upper leaf surface	Lower surface	Sunken stomata
Stomatal behaviour	Din raat khulay	Din mein khulte hain	Raat mein khulte hain
Cuticle thickness	Almost none	Thinner cuticle	Thickest
Water storage	Zaroorat nahin	Thori si	Ample water, succulents kehlate hain

✓ Khud ko check karo (10 sawal)

Pehle khud jawab socho ya notebook mein likho, phir agle hisse mein check karo.

1. Gas exchange kin do openings se hota hai, aur kis process se?
2. Din mein patta kaunsi gas inhale karta hai aur kaunsi exhale?
3. Compensation point of photosynthesis kya hai, aur kab aata hai?
4. Jaron mein gas exchange kin cells ke zariye hota hai?
5. STEAM activity mein tube 2 (andhera) ka indicator kaunsa rang dikhata hai?
6. Homeostasis ki definition likho.
7. Kitab ne kaunse chaar stressors ginwaye hain?
8. Excretophores kise kehte hain?
9. Ebony trees aur conifer trees excretion kis tareeqe se karte hain?
10. Turgid aur flaccid mein kya farq hai?

Jawab

1. **Stomata** (leaves) aur **lenticels** (stems) se, aur process hai **diffusion**.
2. Inhale **carbon dioxide**, exhale **oxygen**.
3. Jab photosynthesis aur respiration ki rates **barabar** ho jayen aur mahol ke saath **koi net gas exchange na ho**. Ye **dawn aur dusk** par aata hai jab light intensity kam hoti hai.
4. **Root epidermal cells** ke zariye, **soil air** ke saath.
5. **Yellow** (CO₂ barh gayi, kyunke andhere mein sirf respiration chal rahi thi).
6. "Homeostasis in plants refers to the **adjustment or adaptation of plants to varying degrees of environmental temperatures and availability of water and salts in the soil.**"
7. **Drought, salinity, pathogens aur herbivores**.
8. Woh patte jo pigmented compounds se bhar kar peelay ho jate hain aur autumn mein gir jate hain, yani patte jo **organ of excretion** ka kaam karte hain.
9. **Ebony trees**: stem aur branches ke purane xylem mein chemicals jama karte hain, jis se beech ki lakri bohat **dark** ho jati hai. **Conifer trees**: jaron se waste compounds mitti mein chhorte hain, doosre paudhon ke khilaf **chemical weapon** ki tarah.
10. **Turgid**: hypotonic mahol mein paani andar aane se cell phool kar kasa hua. **Flaccid**: hypertonic mahol mein paani bahar jane se cell sukar kar dheela.

Score dekho: 8 se 10 sahi: zabardast, chapter 8 mukammal! 🌟 5 se 7: jo ghalat hue sirf un ka hissa dobara dekh lo. 5 se kam: koi baat nahin, ek dafa aur aaram se parho. 💪

Ye chapter 8 ka aakhri part tha. Ab tumhare paas 8.1 se 8.4 tak sab kuch hai: mineral nutrition, transport, gas exchange aur homeostasis.